MATH244 Final Project Presentation

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Project Introduction

Project Overview and Motivation

Motivation

- Determine if song success is driven by tangible characteristics of the music or if it is based on “unmeasureables” (i.e. marketing, luck, timing, “catchiness”)
    - Using a sample of hit and non-hit songs releases between 2010 and 2019
    - We define success as appearance on the Billboard Hot 100
- Why the 2010s?
    - This period represents formative years for many of us (childhood/teens)
    - The music was really good (some songs are still charting today)
    - Successful song criteria seemed to change a lot over this period

Goal:

- Predict success of 2010s songs based on song characteristics
    - Create a model that is over 50% accurate in differentiating hit songs from non-hit songs

Data Sources

MusicOSet

- An open and enhanced dataset of musical elements (music, albums, and artists) suitable for music data mining
- Explanatory Variables:
  - Spotify audio features
  - Spotify artist genres
- Outcome Variable:
  - Billboard Hot 100 Data

The three datasets we used:

Data Cleaning: artists

`main_genre`

- Very specific/niche
- Searched for keywords to sort them into broader genres instead (‘Rap/Hip Hop’, ‘Pop’, ‘Indie’, etc.) and then the nicher genres we put into ‘Other’
- Genres with less than 100 observations in the combined dataset we reclassified into ‘Other’ (prevent overfitting)

Data Cleaning: hits and nonhits

- Drop NA values and missing data
- Convert columns into correct format (numeric or char)
- Drop rows where artist not in artists.csv (can't retrieve genre)
- Drop songs outside of 2010-2019 scope
- Drop >200 artists with incorrect release dates (also drop songs with "remaster" in name)
- Drop songs by artists that didn't have a hit in 2010-2019
- Drop songs with popularity <= 1 (indicates old song uploaded late)
- Equal number of hits and nonhits (random sample)
- Normalizing `loudness`, `tempo`, and `duration_ms`
- Adding outcome variable: `hit` (boolean/logical)
- Joined hits/nonhits into `combined_music_data`
  - Size: 7,392 (3,696 hits + 3,696 nonhits) x 16

Exploratory Data Analysis

Hit Rate by Genre (combined dataset)

Below we can see the hit rate by genre, indicating that certain genres will be more useful than others in predicting success!

```
# A tibble: 12 × 4
   genre                     n n_hits hit_rate
   <chr>                 <int>  <int>    <dbl>
 1 Soundtrack/Theater      914     84     9.19
 2 Country                7923    584     7.37
 3 Soul/R&B/Funk          1812    109     6.02
 4 Hip Hop/Rap           11471    610     5.32
 5 Pop                   25216   1305     5.18
 6 Electronic/Dance       2833    104     3.67
 7 Rock/Metal             9900    344     3.47
 8 Indie                  4716    104     2.21
 9 Other                  6548    143     2.18
10 Easy Listening        18189    281     1.54
11 Comedy/Entertainment   4548     16     0.35
```

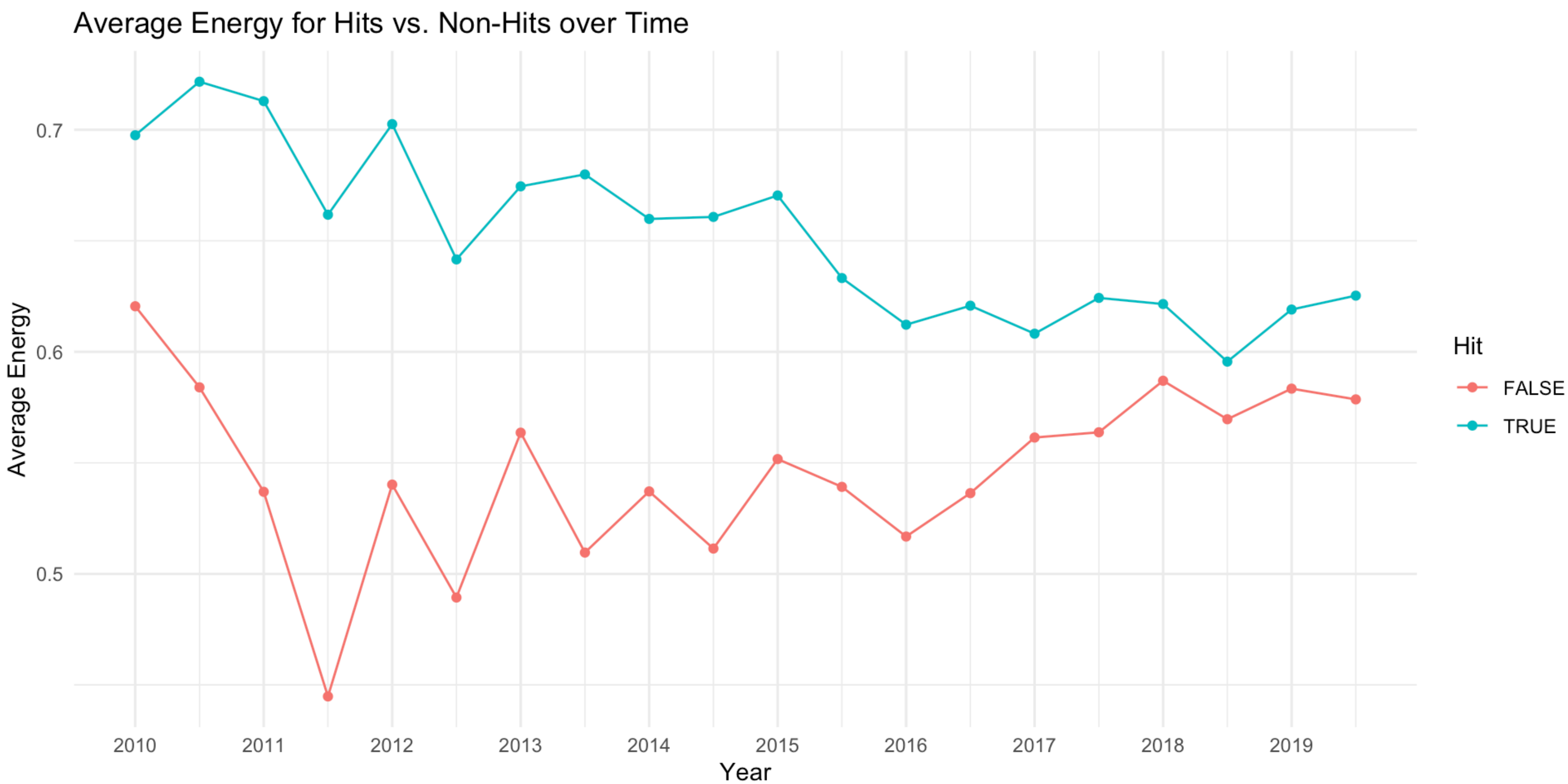
Summary Statistics

Below is a table of summary statistics of our combined dataset for our explanatory variables.

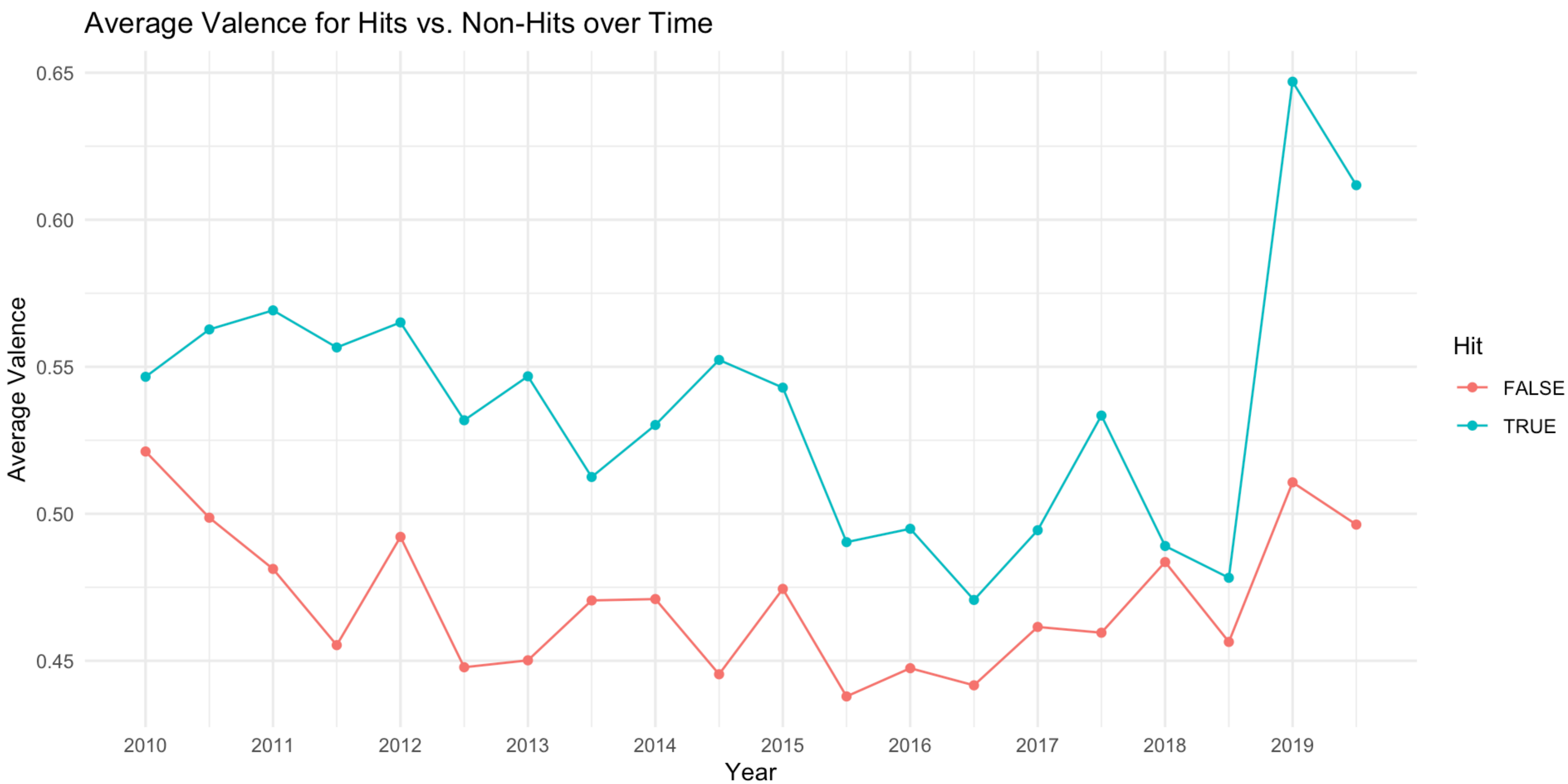
```
# A tibble: 11 × 4
   variable          mean median standard_deviation
   <chr>            <dbl>  <dbl>              <dbl>
 1 duration_ms      0.047  0.045              0.019
 2 acousticness     0.384  0.251              0.362
 3 danceability     0.573  0.585              0.156
 4 energy           0.546  0.561              0.254
 5 instrumentalness 0.185  0                  0.336
 6 liveness         0.217  0.126              0.208
 7 loudness         0.822  0.845              0.087
 8 speechiness      0.089  0.044              0.128
 9 valence          0.471  0.449              0.244
10 tempo            0.489  0.486              0.12
11 explicit         0.117  0                  0.321
```

- Key Takeaways:
  - These statistics are post-normalization
  - Explicit is binary: ~12% of our songs are listed as explicit
  - The distribution of duration is very tight (most songs released at this time had similar lengths)

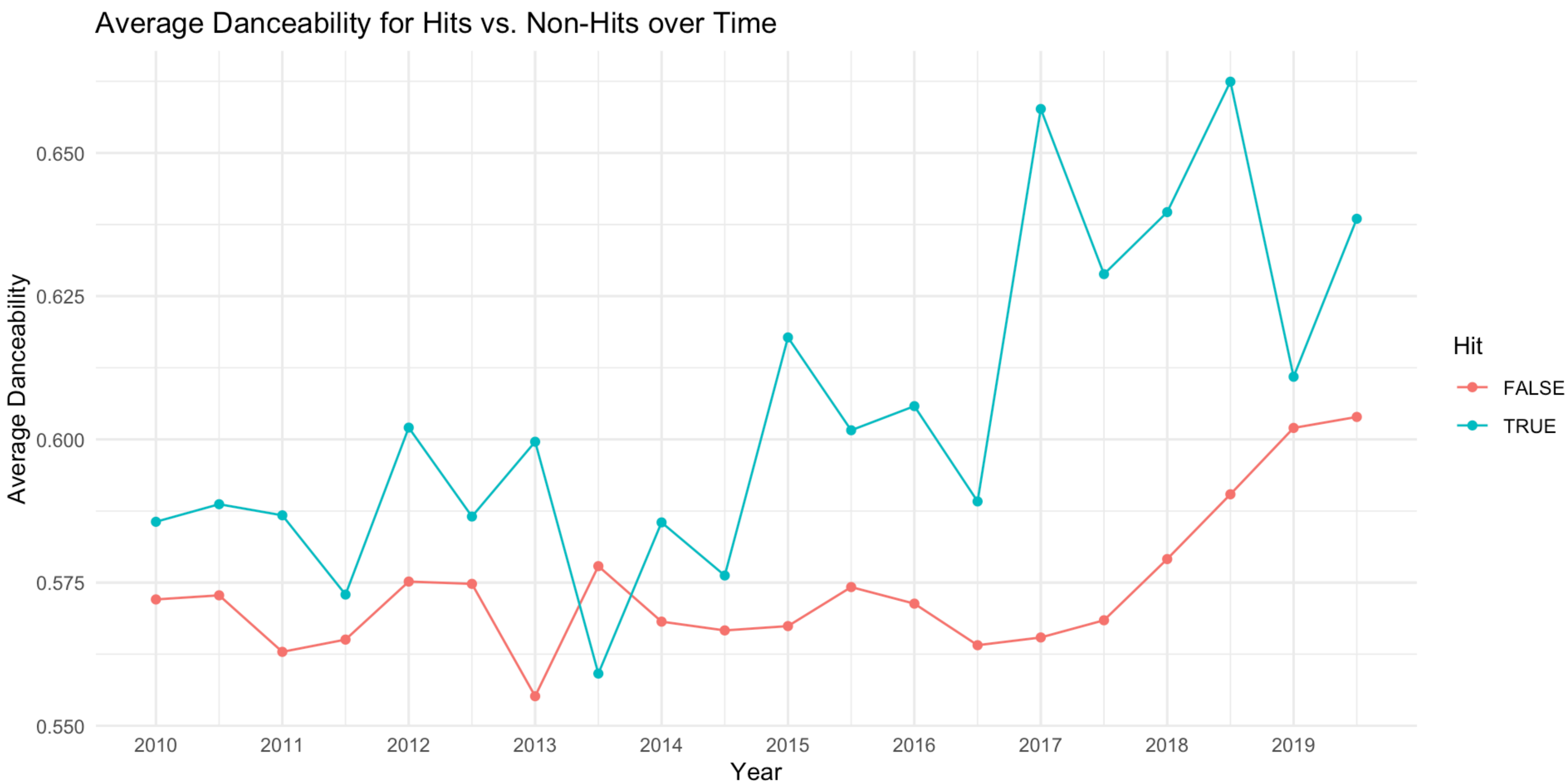
Trends in Energy and hit status



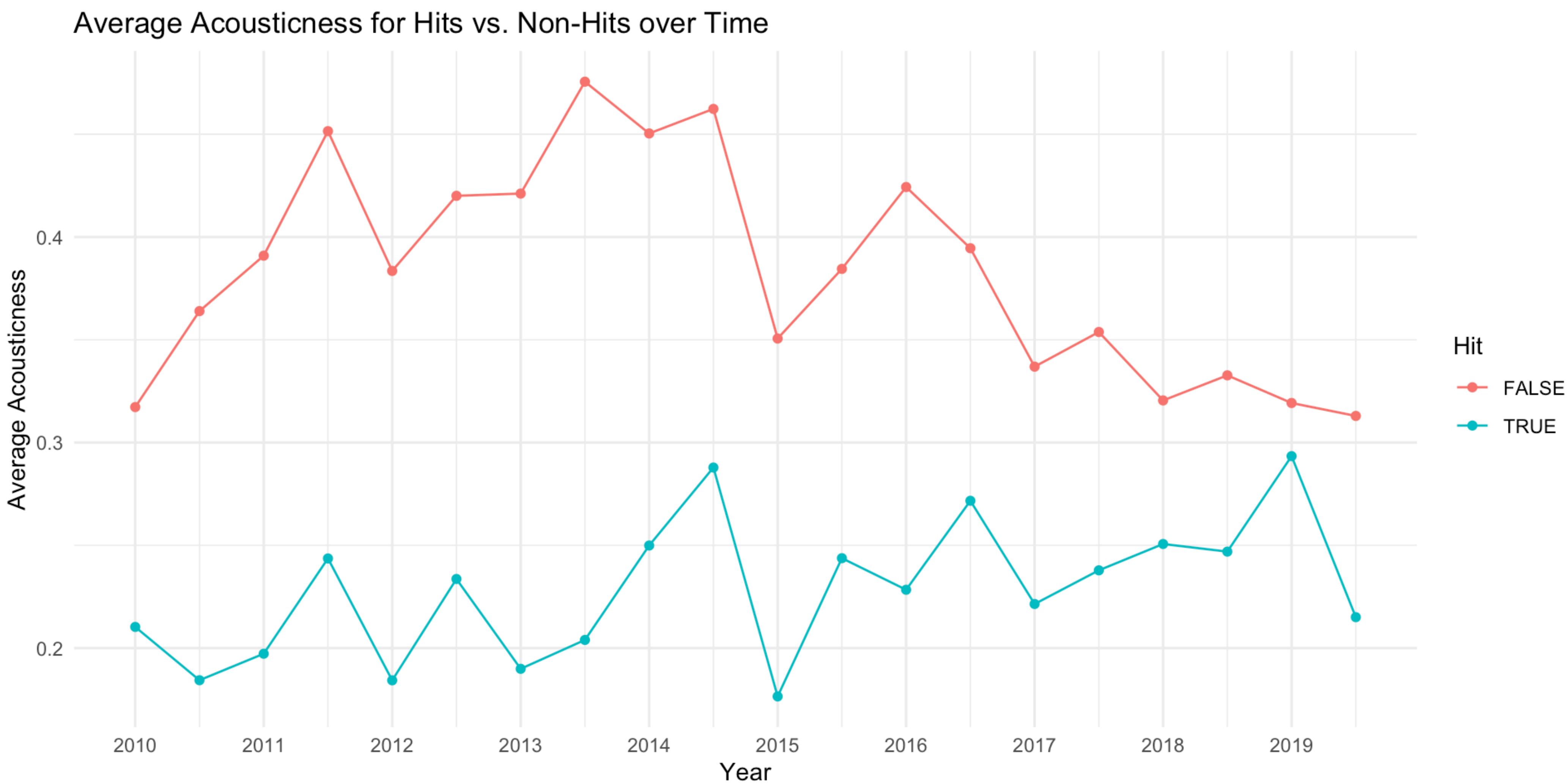
Trends in Valence and hit status



Trends in Danceability and hit status



Trends in Acousticness and hit status



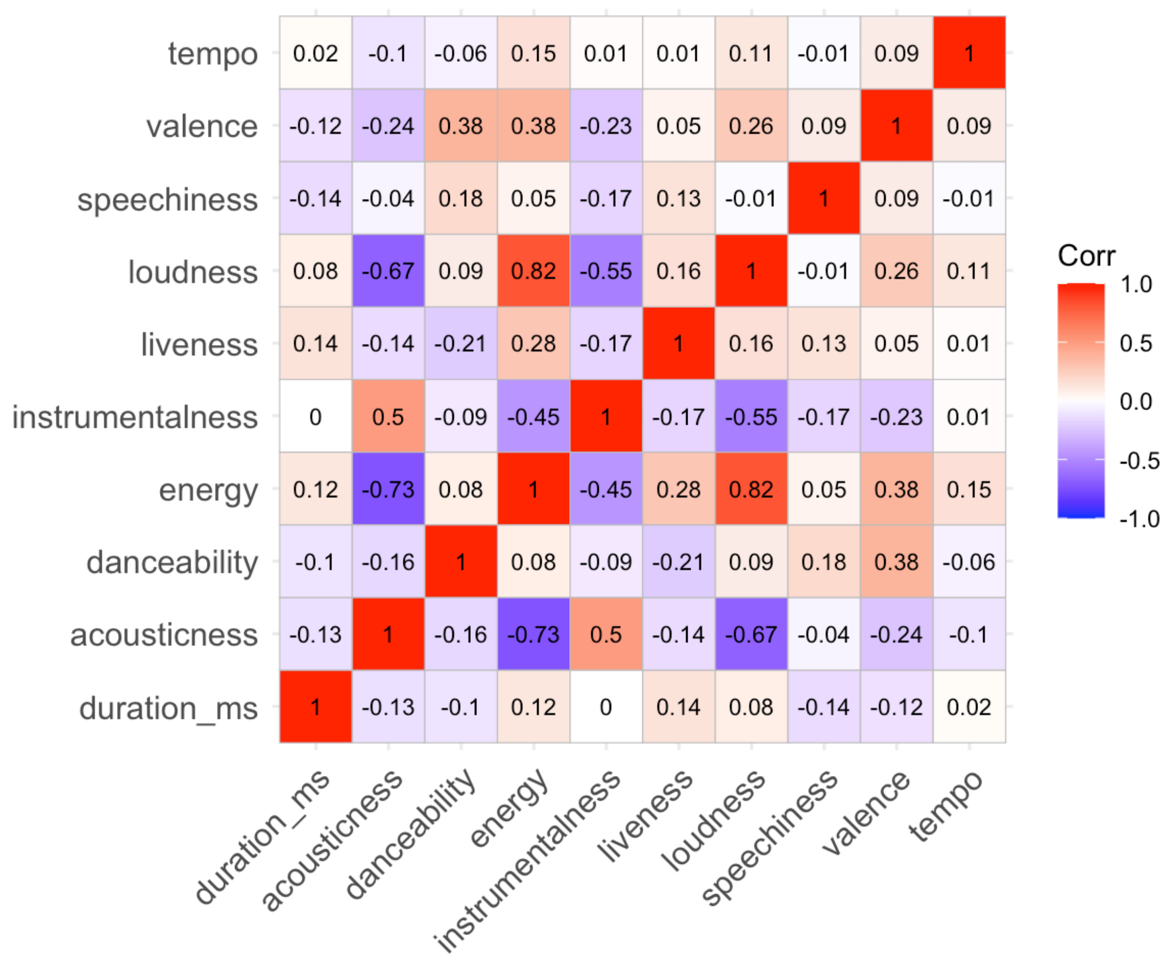
Hypotheses

Hypotheses:

- Song elements that will significantly increase the model's likelihood of predicting a hit:
  - Higher `danceability`, `valence` and `loudness`
  - Lower `duration_ms`, `acousticness`, `speechiness`, `liveness` and `instrumentalness`
- Song elements that will have little to no effect:
  - `key`, `tempo`, `explicit`, and `mode`

Variable Correlation

We made the following image that demonstrates the correlation between the relevant explanatory variables in our dataset.



Inference and Modeling

LASSO and Ridge Classification Models

- Penalizing variables (dealing with multicollinearity)
- Easy to interpret coefficients
- Prevents overfitting (higher bias, lower variance)
- k-fold cross validation: $k = 5$

LASSO coefficient results

	Feature	Lasso
13	genre_Comedy/Entertainment	-1.736152
23	genre_Soundtrack/Theater	1.272827
14	genre_Country	1.133742
15	genre_Easy Listening	0.984211
22	genre_Soul/R&B/Funk	0.961448
20	genre_Pop	0.912735
21	genre_Rock/Metal	0.784560
28	instrumentalness	-0.656785
11	explicit_True	0.655781
17	genre_Hip Hop/Rap	0.642552

LASSO Accuracy results

```
Accuracy score: 70.11%
F1 Score: 0.73
ROC-AUC Score: 0.76
```

Ridge coefficient results

	Feature	Ridge
23	genre_Soundtrack/Theater	2.381823
14	genre_Country	2.180659
22	genre_Soul/R&B/Funk	2.038398
15	genre_Easy Listening	2.004224
20	genre_Pop	1.935171
21	genre_Rock/Metal	1.812181
17	genre_Hip Hop/Rap	1.667668
16	genre_Electronic/Dance	1.491713
19	genre_Other	1.335669
18	genre_Indie	1.118170

Ridge Accuracy results

```
Accuracy score: 70.52%

F1 Score: 0.74

ROC-AUC Score: 0.76
```

LASSO coefficient results (no genre)

	Feature	Lasso
17	instrumentalness	-0.654009
11	explicit_True	0.552427
19	loudness	0.448967
1	key_2.0	-0.218562
20	speechiness	-0.184922
18	liveness	-0.148007
21	valence	0.147529
7	key_8.0	-0.092039
0	key_1.0	-0.086384
9	key_10.0	-0.070223

LASSO Accuracy results (no genre)

```
Accuracy score: 66.4%
F1 Score: 0.71
ROC-AUC Score: 0.73
```

Ridge coefficient results (no genre)

	Feature	Ridge
17	instrumentalness	-0.635130
11	explicit_True	0.492469
19	loudness	0.432877
1	key_2.0	-0.203961
20	speechiness	-0.173074
18	liveness	-0.148231
21	valence	0.137684
7	key_8.0	-0.092181
12	mode_1.0	0.084166
0	key_1.0	-0.083744

Ridge Accuracy results (no genre)

```
Accuracy score: 66.4%
F1 Score: 0.7
ROC-AUC Score: 0.73
```

Discussion and Future Work

Conclusions + Limitations

- Model does not *fully* capture what makes a song a hit, but it does a pretty decent job
  - Achieved around 70% accuracy
- Genre is highly influential
  - Model performance increases by ~3% when genre is included
  - Spotify-derived characteristics are not as important
- Dates may not reflect reality
  - Unlabeled remasters released during the sample period may bias results
    - These songs may not follow the 2010s “formula for success”
  - Certain songs have incorrect dates due to when they were added to music streaming platforms
  - Potential use of Spotify API to get correct dates
- Varying amount of data by year/genre
  - Different numbers of observations per year/genre may also be skewing data

Future Work

- Look further into genres
  - Certain genres with low numbers of observations may be biasing our current estimates
- Test model on currently successful songs/new releases
  - See how accurate the model is with music from outside this period
- Potentially looking at other predictors
  - Collaborations vs. Solo Song
  - Whether the artist is signed vs. independent
    - Could help proxy some of the “marketing” impact
  - A song’s language

Thank you :)